

Tool	URL
Trifacta	https://www.trifacta.com/start-wrangling/
RapidMiner	https://rapidminer.com/
Rattle	https://cran.r-project.org/bin/windows/base/
QlikView	http://global.qlik.com/us/landing/go-sm/qlikview/download-qlikview
Weka	https://www.cs.waikato.ac.nz/ml/weka/
KNIME	https://www.knime.org/knime-analytics-platform
Orange	http://orange.biolab.si/
Tableau Public	https://public.tableau.com/s/
Open Refine	https://openrefine.org/
Talend	https://www.talend.com/products/data-preparation/#free-desktop
Data Preparator	http://www.datapreparator.com/downloads.html
Tanagra	http://eric.univ-lyon2.fr/~ricco/tanagra/en/tanagra.html
H2O	http://www.h2o.ai/download/h2o/choose
Gephi	https://gephi.org/

Table 1: Commonly used tools for data exploration and analytics

The following are the key functionalities in Python Pandas:

- Effective DataFrame management for data sets
- Loading and processing of data in multiple formats
- Reshaping of data for assorted research implementations
- Pivot table generation and visualisation
- Grouping and aggregation of data
- High-performance data wrangling
- Missing value imputations and data treatments
- Indexing, slicing and subsetting of large data sets
- Melting, joining, concatenation and merging
- Time series based analysis and predictions

Data analysis on a COVID-19 data set using Python Pandas

Currently, the world is facing the menace of COVID-19 that has resulted in the loss of millions of lives worldwide. Medical experts as well as data scientists are making ceaseless efforts to analyse

	A	B	C	D	E	F	G	H	I
1		SNo	ObservationDate	State	Country/Region	Last Update	Confirmed	Deaths	Recovered
2	9	10	01/22/2020	Hebei	Mainland China	1/22/2020 17:00	1	0	0
3	47	48	01/23/2020	Hebei	Mainland China	1/23/20 17:00	1	1	0
4	111	112	01/24/2020	Hebei	Mainland China	1/24/20 17:00	2	1	0
5	146	147	01/25/2020	Hebei	Mainland China	1/25/20 17:00	8	1	0
6	190	191	01/26/2020	Hebei	Mainland China	1/26/20 16:00	13	1	0
7	237	238	01/27/2020	Hebei	Mainland China	1/27/20 23:59	18	1	0
8	286	287	01/28/2020	Hebei	Mainland China	1/28/20 23:00	33	1	0
9	336	337	01/29/2020	Hebei	Mainland China	1/29/20 19:30	48	1	0
10	389	390	01/30/2020	Hebei	Mainland China	1/30/20 16:00	65	1	0
11	448	449	01/31/2020	Hebei	Mainland China	1/31/2020 23:59	82	1	0
12	509	510	02-01-20	Hebei	Mainland China	02-01-20 1:52	95	1	0
13	577	578	02-02-20	Hebei	Mainland China	2020-02-02T02:53:02	104	1	3
14	645	646	02-03-20	Hebei	Mainland China	2020-02-03T00:23:14	113	1	3
15	712	713	02-04-20	Hebei	Mainland China	2020-02-04T08:53:01	126	1	4
16	782	783	02-05-20	Hebei	Mainland China	2020-02-05T11:23:18	135	1	6
17	853	854	02-06-20	Hebei	Mainland China	2020-02-06T13:53:02	157	1	13
18	924	925	02-07-20	Hebei	Mainland China	2020-02-07T12:03:04	172	1	22
19	994	995	02-08-20	Hebei	Mainland China	2020-02-08T11:33:02	195	1	30

Figure 1: Data set of COVID-19 with date-wise records

the patterns of COVID-19 virus using different techniques.

The data sets of COVID-19 are available on the public domain for researchers and academicians, so that solutions and predictions can be worked out by different experts.

Listed below are a few of the links to data sets on COVID-19 which are publicly available for data scientists and researchers:

- data.humdata.org/dataset
- ieee-dataport.org/open-access/corona-virus-covid-19-tweets-dataset
- data.world/datasets/covid-19
- kaggle.com/sudalairajkumar/novel-corona-virus-2019-dataset
- kaggle.com/imdevskp/corona-virus-report
- github.com/datasets/covid-19
- dimensions.ai/news/dimensions-is-